

Database Management Systems - Lecture Notes

1. Introduction to DBMS

A Database Management System (DBMS) is software used to create, store, organize, retrieve and manage data efficiently. Examples include MySQL, PostgreSQL, Oracle and Microsoft SQL Server.

2. Database Models

Common database models include hierarchical, network, relational and object-oriented models. The relational model stores data in tables consisting of rows and columns.

3. Normalization

Normalization is the process of organizing data to reduce redundancy and improve data integrity. First Normal Form (1NF) requires atomic values. Second Normal Form (2NF) removes partial dependencies. Third Normal Form (3NF) removes transitive dependencies.

4. Primary and Foreign Keys

A primary key uniquely identifies each record in a table. A foreign key creates a relationship between two tables by referencing a primary key in another table.

5. SQL

SQL is used to interact with relational databases. Important commands include SELECT, INSERT, UPDATE and DELETE. The SELECT statement is commonly used to retrieve records from a table.